Rahul Kaushal

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Technical Skills

Languages & Backend: Python, Go, SQL, Bash, FastAPI, asyncio, Pydantic, SQLAlchemy, REST/WebSocket APIs, pytest, Behave (BDD), Locust
Data & ML: PySpark, Databricks, Delta Lake, Spark Structured Streaming, PostgreSQL, PGVector, MLflow, LangChain, LLM/RAG, Azure OpenAI, Vertex AI
Infrastructure: Kubernetes (MicroK8s, AKS), Docker, Ansible, Ceph, Pacemaker/Corosync, Azure (Event Hub, IoT Hub, Blob Storage), GitHub Actions, Prometheus/Grafana

Professional Experience

Advanced Software Engineer *May 2026 – Present*
Honeywell

- Contributed to the performance work that cut incremental-update install time 32.5% (averaged 90m11s to 60m51s on a 44-package run). The biggest cost was a fixed datastore-drain pause; no package's install script came close. We consolidated 13 sequences into 9, halved the pause, and moved extraction to `pigz`.
- Worked on EdgeOrchestrator, the team's Go control-plane service, working on its REST/WebSocket API and the Ansible-driven deploy and upgrade path over the MicroK8s, Ceph and Pacemaker stack. Playbooks run as a subprocess with an argument array (nothing reaches a shell), and a marker file stops a crashed self-update from reporting success.
- Helped root-cause a watchdog feedback loop wedging idle appliances. A hard MicroK8s kill left Calico IPAM allocations unreleased until the address pool ran out, and the watchdog's restart re-triggered it. Fixed with registry image pre-caching, a Recreate rollout for the single-writer volume, and a circuit breaker halting auto-restarts after 5 escalations in 24h.
- Collaborated on ForgeLinkLite, a cross-platform Go agent handling Azure DPS enrollment and IoT Hub connectivity over AMQP-over-WebSocket with certificate auth, a KeePass credential vault, and active/passive handshake failover.

Go, Kubernetes (MicroK8s), Ansible, Ceph, Pacemaker/Corosync, Azure IoT Hub

Software Engineer II *Jul 2024 – Apr 2026*
Honeywell

- Built the deployment path routing models to Databricks Serving, AKS or edge devices from one API, with discriminated-union request types validating each target's config at the boundary and status tracked by two independent asyncio tasks.
- Wrote the asyncio Event Hub processor: three parsers behind a Factory Pattern, concurrent PostgreSQL and Unity Catalog writes, and Blob checkpoints committed only after a successful write. Delivery is at-least-once with upsert writes. A crash replay overwrites rows without duplicating them.
- Routed model registrations through five security scanners (Coverity, Black Duck, Twistlock, ModelScan, AST parser) behind one Strategy Pattern interface. Adding a sixth would mean a new class, not another conditional.
- Fine-tuned Gemini on Vertex AI to translate SkySpark Axon rules into Honeywell's Forge DSL, with a versioned prompt store and tuning-job state machine. A translation counted as correct only if it produced the same results as the original rule, checked by the domain experts who supplied the training pairs.
- Shipped the SkySpark migration service (SCRAM auth, Zinc/Trio parsing, parallel writes that roughly halved write time) and the team's Behave suite of 50+ scenarios across six workflow domains, run in parallel per tenant, plus Locust load tests.

Python, FastAPI, Databricks, Azure Kubernetes (AKS), Google Vertex AI, MLflow, Azure Event Hub, PostgreSQL

Data Engineer II *Jun 2023 – Jun 2024*
Honeywell

- Contributed to 15+ PySpark ingestion pipelines on Databricks across 12+ source systems (SQL Server, Salesforce, SharePoint, field-service databases) into Delta Lake, using timestamp watermarking for incremental loads.
- Engineered a Spark Structured Streaming pipeline over PDF arrivals in Blob Storage using foreachBatch to run a per-micro-batch Delta MERGE INTO for SCD Type 2 upserts, reconciliation against processed records to catch source deletions, and Change Data Feed to let the vectorization job read only changed rows.
- On the Maintenance Assist chatbot, worked on the agent executor (tool selection, chat history, summarization) with a YAML config that let each tenant run its own LLM and prompts without a code change. Also helped build async guardrails and a PGVector memory store that retrieves turns by max inner product similarity.
- Wrote the SQL transformations for field service analytics (incident resolution trends, contract and maintenance metrics, NPS) that feed the customer-facing DSP dashboard.

PySpark, Databricks, Delta Lake, PostgreSQL, LangChain, PGVector, Azure OpenAI, MLflow

Patent

Data management in a multi-tenant environment | US 12,717,808 *Issued Aug 2026*
Tenant isolation at the data level on a shared multi-tenant backend: records carry tenant metadata and are split and routed by customer, product or a custom rule, instead of a separate schema or database per tenant.

Open Source

DepKeeper | *Python, Click, httpx, asyncio* *2025 – Present*
A CLI that upgrades `requirements.txt` without crossing a major version boundary, preserving your upper bounds and exclusions. Every change previews as a dry run first, and re-running it is idempotent. On PyPI under Apache 2.0.

Education

B.Tech Biotechnology + M.Tech Data Science *2018 – 2023*
Indian Institute of Technology Madras *CGPA: 8.77*